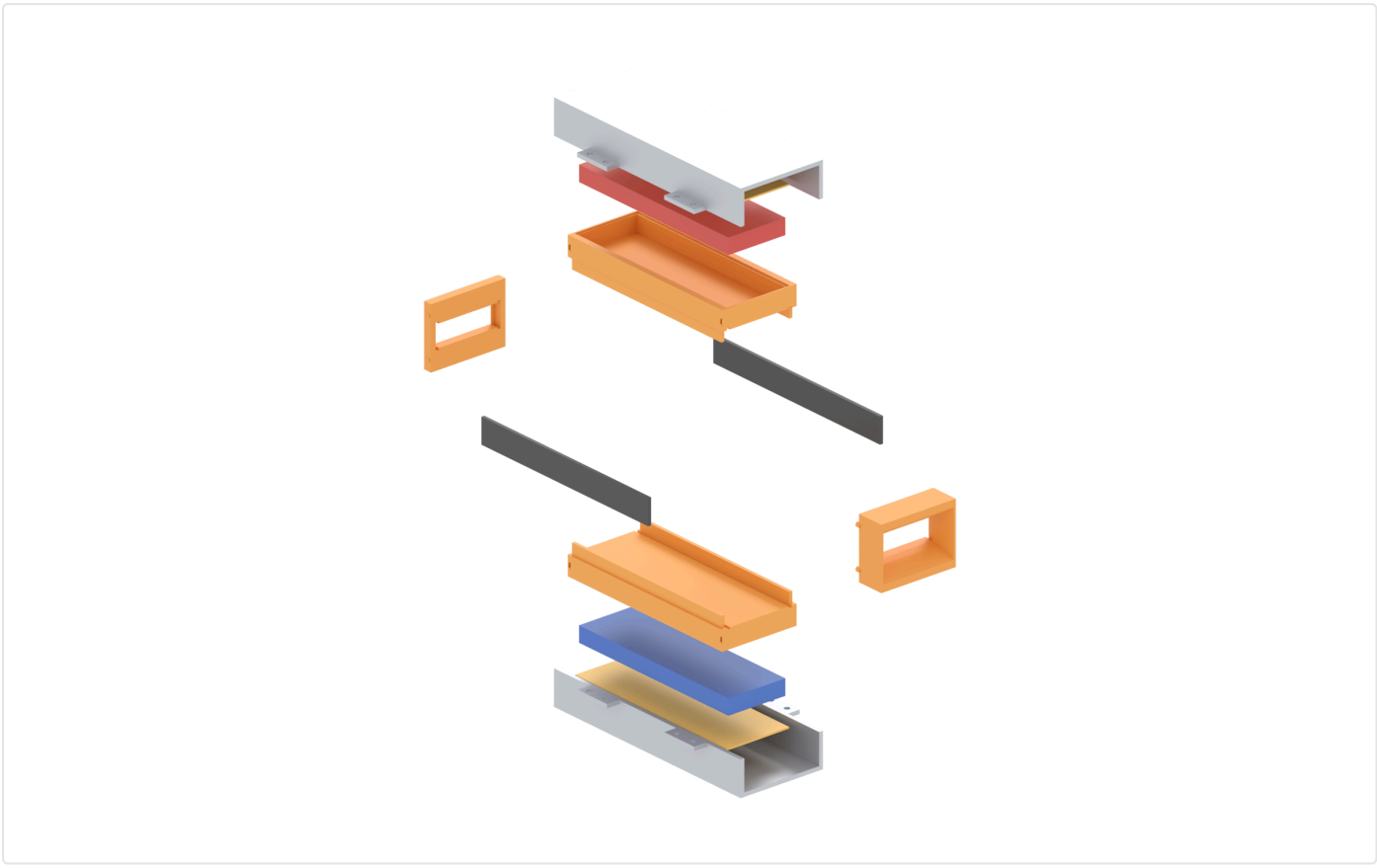
Magnetide v21-r2h — Assembly Instructions

Reynard Filbert Tang · Kelas 10.8, Kelompok 1 · MHD seawater pump
Envelope 150 × 72 × 44 mm · bore 40 × 14 × 124 mm · all dimensions in mm · rev r2h



Exploded. Upper half, bottom to top: yoke, tray with its magnet cavity, four magnet bars, 1 mm cap. Bellmouth left, rear block right, electrodes outboard.

What you are building

Brine runs through a 40 × 14 mm bore. A 483 mT field crosses it, current flows between two graphite plates on the side walls, and the Lorentz force pushes the water out the straight end. The steel yoke is the magnetic return path; the plastic tray is the only thing keeping water away from the magnets.

Stack, per half

z	Layer	Thickness	Function
0 – 7	bore	14 tall	water channel, 40 wide
7 – 8.5	tray floor	1.5	only barrier between water and magnets
8.5 – 18.5	magnet cavity	10	closed on all six sides
18.5 – 19.5	cap	1.0	clamps the stack down, flush with steel
19.5 – 22	steel plate	2.5	flux return

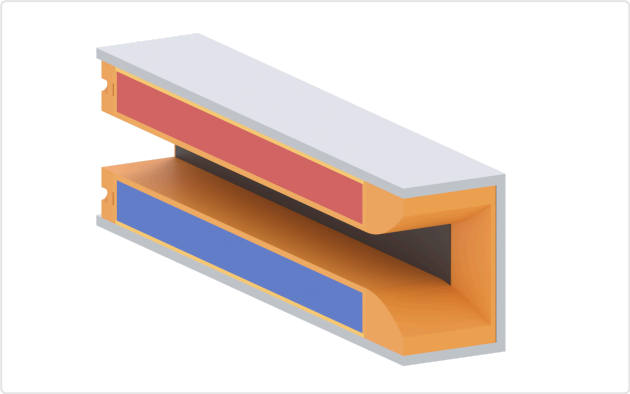
Build sequence

1 Fold the two yokes

Cut two blanks from `laser/clamshell-flat-2.5mm.dxf`. Fold 90° down at ±28 (the walls), then fold the two lugs out 90° at ±47.96.

Check: 56 mm across the plate, 22 mm from plate face to lug face, **two Ø3.4 holes per lug** (13 mm apart) at x 40 and x 110.

2 Stack the magnets into the tray cavity



Eight tiles per row, four rows, two layers deep — 48 per face. Two 5 mm layers make the 10 mm bar and fill the cavity 8.5 → 18.5.

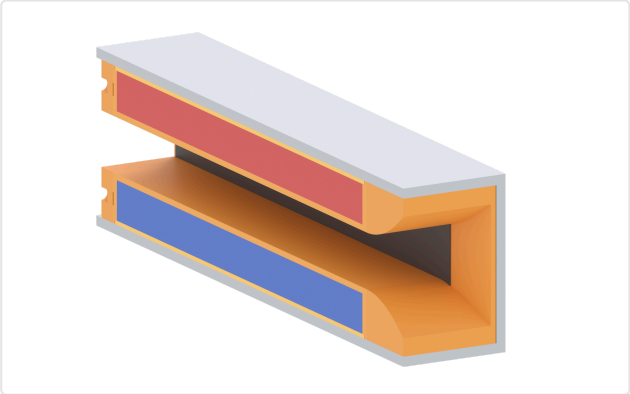
Compass-check every row before the epoxy sets. All four rows on one face point the same way; the opposite face points the other way. One flipped row stalls that quarter of the bore.

> Tiles are modelled as 9.7 mm bars with 0.3 mm clearance so they drop in. Do not press-fit — a cracked tile corrodes from the crack.

3 Cap it

Drop the 1 mm cap into the ledge in the tray rim. It sits on the magnets with its top face flush with the steel inner face, and stops the stack lifting when the field pulls. A bead of epoxy at the rim; never a puddle — epoxy in the bore is unreachable.

4 Seat the electrodes



Each 134 × 14 × 2 plate slides into the 1 mm groove from the end. Its top shoulder hides behind 0.5 mm of floor; the full 14 mm face is exposed — 124 × 14 = **1736 mm²** per electrode.

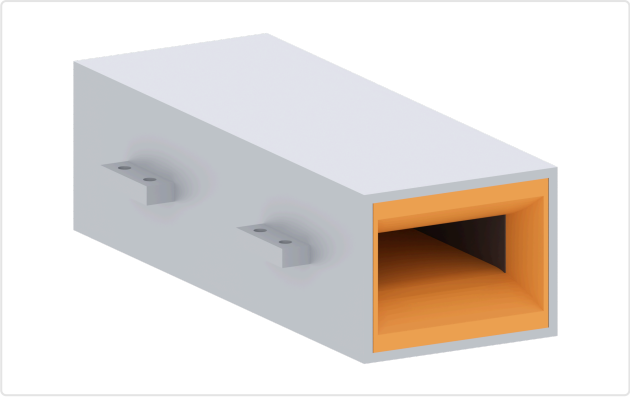
Push each plate until its end sits 5 mm inside the bellmouth or rear-block pocket. That pocket is the end seal.

5 Bellmouth and rear block

Two 2 mm pegs each, into the blind holes in the tray end faces. The bellmouth takes the 48 × 30 mouth down to the bore over 20 mm.

At the outlet, drill through the last 0.4 mm of the lead channel, feed the two wires out, epoxy the hole closed around them. Keep the outlet straight.

6 Close the yoke



Lower tray into the bottom yoke, upper into the top, bring the halves together so the lug faces meet. Eight M3 through the lugs — two per lug, 13 mm apart — tightened in a cross pattern a little at a time.

This is a **stack clamp**: the tray rim lands on the steel plate above, the mirrored tray butts it at z 0, and the bolts close the sandwich. Sideways there is 0.15 mm a side of clearance and nothing clamps it.

> **If the tray rattles sideways**, shim it — closed-cell foam strip 0.3–0.5 mm on the *outer wall faces* only, outboard of the tray. Never between cap and plate: that face is the magnet's path into the steel, and foam there is an air gap in series with every magnet. Never open-cell foam, which wicks brine.

Watertight: the three paths that were open

Leak path	What was wrong	Fix in r2h
electrode groove roof	2.2 mm groove cut 3.0 deep from z 7.0, so its ceiling sat at z 10.0 — **1.5 mm above the cavity floor at 8.5**. A gutter into the magnet pocket.	groove 1.0 deep (7.0 → 8.0), ceiling 0.5 mm below the floor
end-block peg holes	holes 2.6 mm deep in a 2.0 mm wall — through-holes into the cavity at both ends	blind 1.6 deep, 0.4 mm of plastic behind, 2.3 mm lead-in
magnet lead exit	full-length bore opened the cavity to the outside	blind channel, 0.4 mm left; drill and epoxy at wiring time

Before you power it

- ☐ Both halves flush, 8 bolts seated in the lugs, no gap at the lug faces
- ☐ Compass: every row on a face points the same way, the two faces oppose
- ☐ Tray floor unbroken over the whole magnet area — hold it up to a light
- ☐ Electrodes fully home, not shorting to steel through a scratch
- ☐ Lead exit drilled, wired, epoxied; bore clear down its whole 124 mm
- ☐ Fuse in the supply line; runs under 2 minutes, ventilated (Cl_2)